EvoByte: Massively Parallel Evolutionary Discovery of Symbolic Programs via Ultra-Compact Bytecode and Deterministic Cascade Verification

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Abstract

Symbolic regression searches for closed-form expressions that fit observations. We present **EvoByte**, a program-synthesis engine built on a fixed 64-byte linear bytecode virtual machine and a four-stage verification cascade (S0–S3) designed for GPU execution without string or AST parsing on the hot path.

We evaluate the falsifiable Hypothesis H1 on two Level-A targets ($x+1$ and x^2+3x+7) with wall-clock budgets of 0.4 s and 0.8 s on an NVIDIA GeForce RTX 4060 Laptop GPU against random search, a classic genetic-programming baseline and a classical symbolic-regression baseline; the PySR adapter was not executed (missing dependency) and is reported as not run. The cascade yields a measured 15.51× throughput speedup over the project's naive full-dataset evaluation, and early termination a further 3.00×. Evolutionary search beats random search (Wilcoxon $W = 55.0$, Holm-adjusted $p = 2.49 \times 10^{-3}$), but the classical symbolic-regression baseline reaches 100% success versus 80% for EvoByte, with lower hidden error and higher throughput. H1's honest-baseline criterion is therefore *not supported*. We report this negative result together with all raw artifacts and reproduction commands. A P161–P170 validation program re-audited every claim: a mechanical integrality audit retracted 49 of 58 Schinzel identities, a Lean statement-fidelity gate found 2063 provably vacuous statements out of 4888, a preregistered 120-trial combinatorial record campaign (12,016,204 evaluations) found no new record, and the consolidated statistical verdict is H1 *not supported* with *null result* for mathematical record discovery.

1 Introduction

Automated discovery of symbolic mathematical equations from observational data constitutes a foundational challenge in computational scientific discovery [10, 11]. Unlike black-box neural networks, symbolic mathematical equations offer interpretability, parsimony, and extrapolation fidelity grounded in physical and mathematical principles [2].

For over three decades, Genetic Programming (GP) has served as the dominant paradigm for symbolic discovery [5, 9]. Standard GP represents candidate expressions as variable-sized Abstract Syntax Trees (ASTs). While expressive, tree structures exhibit fundamental computational liabilities when deployed on modern parallel architectures:

1. **Tree Bloat:** Evolutionary operators inherently inflate tree depth and node count without corresponding gains in fitness, consuming memory and degrading interpretability [9].

2. **Memory Fragmentation & Pointer Chasing:** AST evaluation requires non-contiguous heap allocations and recursive tree traversals, causing severe cache thrashing on CPU caches and low SIMD utilization on GPUs [6].

3. **High Verification Overhead:** Scoring large candidate populations against multi-point numerical grids is computationally prohibitive when evaluated in a monolithic pass.

To overcome these structural limitations, we formulate and systematically test **Hypothesis H1**: *In problem domains where candidates can be verified rapidly, massively parallel evolutionary search over ultra-compact, fixed-size linear bytecode coupled with a hierarchical verification cascade matches or outperforms traditional tree-based algorithms per unit of wall-clock compute time.*

In this paper, we present the architecture, empirical evaluation, and reproducibility infrastructure of **EvoByte**. The core contributions of this work are:

- **Fixed-Width Bytecode Architecture:** A 64-byte linear virtual machine operating over 8 registers with zero dynamic memory allocation and vector-parallel execution in GPU VRAM.
- **Hierarchical Verification Cascade (S0–S3):** A multi-tier screening engine that rejects invalid or inaccurate candidates early, achieving a measured 15.51× speedup over monolithic evaluation.
- **Statistically Rigorous Multi-Seed Validation:** Strict wall-clock benchmarking confirming the advantage of evolution over *random search* via Wilcoxon signed-rank testing with Bonferroni–Holm FWER control ($p \leq 0.01$); the classical symbolic-regression baseline is not outperformed.
- **Generalization & Anti-Bloat Properties:** EvoByte reaches an 80.0% extrapolation rate, 0.0% asymptotic explosion rate, and average model size of 5.2 opcodes on hidden and extrapolation splits.
- **Comprehensive Ablation Battery:** Systematic measurement of all nine architectural components, including the empirical rejection (DROP) of neural micro-generators in accordance with measured efficiency.
- **Independent Hermetic Reproducibility:** A fully containerized execution environment operating under '–network none' and an open-access research bundle with cryptographic SHA-256 manifests.

2 EvoByte Virtual Machine Architecture

2.1 Linear Bytecode Specification

Rather than dynamic trees, EvoByte encodes candidate programs as contiguous byte arrays of fixed size (64 bytes). Each instruction occupies an 8-bit opcode followed by register and immediate operands:

$$I_k = \langle \text{OPCODE}, r_{\text{dest}}, r_{\text{src1}}, r_{\text{src2}} \rangle \tag{1}$$

The virtual machine provides 8 general-purpose floating-point registers ($r_0, \ldots, r_7$), with r_0 initialized to input variable x and $r_1, \ldots, r_7$ initialized to constants or zeros. The instruction set comprises elementary arithmetic and transcendental primitives:

$$\Omega = \{\text{NOP}, \text{ADD}, \text{SUB}, \text{MUL}, \text{DIV}, \text{SIN}, \text{COS}, \text{EXP}, \text{LOG}, \text{SQRT}\} \tag{2}$$

2.2 Protected Arithmetic Without Warp Divergence

On GPU hardware, branch divergence within a thread warp causes serialized execution. EvoByte eliminates branching by implementing branchless protected operators using tensor masks:

$$\text{div}(a, b) = \frac{a}{b + \epsilon \cdot \text{sgn}(b)} \cdot \mathbf{1}_{\{|b|>\epsilon\}} + 1.0 \cdot \mathbf{1}_{\{|b|\leq\epsilon\}} \tag{3}$$

$$\log(a) = \ln(|a| + \epsilon) \tag{4}$$

$$\text{sqrt}(a) = \sqrt{|a|} \tag{5}$$

All intermediate register values are clamped to finite bounds $[-10^6, 10^6]$, preventing floating-point overflow and IEEE NaN/Inf propagation across parallel threads.

2.3 GPU-Resident Population State

The entire evolutionary population is represented as a contiguous 2D integer tensor $P \in \mathbb{Z}^{N\times 64}$, where N denotes population size ($N \in [50, 1000]$). During evolution, point mutations, uniform crossovers, and register allocations are computed directly on GPU tensors using PyTorch vector operations. Candidate expressions never transit between host CPU and device GPU memory during the inner evolutionary loop.

3 Hierarchical Verification Cascade (S0–S3)

A critical bottleneck in symbolic regression is candidate fitness evaluation. Evaluating every generated program across M evaluation points requires $O(N \cdot M)$ operations. EvoByte introduces a four-stage hierarchical verification cascade:

1. **Stage S0 (Static Validation):** Verifies syntactic integrity, register index bounds, non-trivial opcode presence, and finite parameter boundaries ($O(1)$ cost per candidate). Infeasible programs are dropped before tensor execution.

2. **Stage S1 (Coarse Grid Screening):** Evaluates surviving candidates on a sparse subgrid of $M_{\text{coarse}} = 32$ points. Programs failing an initial Mean Squared Error (MSE) threshold ($\text{MSE} > \theta_{\text{coarse}}$) are immediately pruned.

3. **Stage S2 (Fine Grid with Early Termination):** Evaluates candidates on the full training grid ($M_{\text{fine}} = 256$ to 512 points). Evaluation accumulates squared errors in batches, terminating execution prematurely if accumulated error exceeds the current best population threshold.

4. **Stage S3 (Exact Symbolic Simplification):** Canonical reduction and algebraic equivalence checking via computer algebra systems and formal Lean 4 kernels [3].

As detailed in Table 1, the 15.51× entry is the measurement of the *complete* multi-stage cascade (ablation A8, "No Cascade", `speedup_ratio` = 15.51) against naive full-dataset scoring. Early termination (ablation A9, "No Early Termination", `speedup_ratio` = 3.00) contributes a further **3.00**× speedup. Executing the same bytecode on the GPU instead of the CPU (ablation A6, "CPU vs GPU Execution", `speedup_ratio` = 2.60) yields a **2.60**× speedup; this ablation measures device execution, not population residency in VRAM. The remaining ablations (A1, A2, A4, A5, A7) show no measured throughput speedup (`speedup_ratio` = 1.00) and were retained on the basis of their recorded qualitative rationales; ablation A3 (neural micro-generator) was dropped under ADR-0010. Under the strict budgets used here EvoByte verified between 500 and 25,017 candidates per run, with mean throughputs of 3,326 CVPS (P154 budget matrix) and 4,800 CVPS (P155 paired baselines) on reference consumer GPU hardware.

Table 1: Measured 9-ablation battery: each architectural component removed once against the full system on the target $y = x^2 + 3x + 7$ (P157, 5 seeds). The **Measured Speedup** column reports the throughput speedup of the full system over the variant without the component (from `experiments/p157-ablations/summary_ablations.json`, field `speedup_ratio`); a value of 1.00× means no throughput speedup was measured for that component, and the keep/drop decision follows the recorded rationale. The 15.51× entry is the complete cascade (A8, executed without the cascade), not a single stage.

Ablation	Component Removed	Measured Speedup	Decision
A1	MAP-Elites & Novelty Fitness	1.00×	**KEEP**
A2	Single-Point Crossover	1.00×	**KEEP**
A3	Micro Bytecode Sampler (ADR-0010)	1.00×	**DROP**
A4	4-Island Ring Migration	1.00×	**KEEP**
A5	Tunable Slots + Local Search	1.00×	**KEEP**
A6	Resident PyTorch GPU VM (CPU instead of GPU)	**2.60**×	**KEEP**
A7	Massive Population Batching	1.00×	**KEEP**
A8	Multi-Stage Verifier Cascade (S1–S3)	**15.51**×	**KEEP**
A9	Early Stop Rejection on Overflow/NaN	**3.00**×	**KEEP**

4 Experimental Protocol

4.1 Falsifiable Hypothesis Criteria

Hypothesis H1 is defined by five falsifiable criteria preregistered in the repository architecture (`docs/HYPOTHESIS.md`):

- **C1 (Rediscovery):** Recovery of Level-A primary targets in $\geq 4/5$ independent seeds within 1 h on 8 GB VRAM. The executed campaigns (P154–P156) used $y = x + 1$ and $y = x^2 + 3x + 7$ under budgets of 0.5 s–1.5 s and 0.4 s–0.8 s; the preregistered $y = \sin(x) + x^2$ target was not part of them.
- **C2 (Speed):** Verifiable cascade speedup $\geq 10.0\times$ and early termination speedup $\geq 2.0\times$.
- **C3 (Value of Evolution):** Statistically significant advantage of genetic search over pure random search under equal wall-clock time ($p \leq 0.01$).
- **C4 (Generalization & Anti-Bloat):** Extrapolation accuracy $\geq 70.0\%$, asymptotic explosion rate of 0.0%, and compact complexity (≤ 25 opcodes).
- **C5 (Honest Baselines & Pareto Frontier):** Equal-budget comparison against Random Search, Classical GP, and PySR, extracting non-dominated Pareto sets.

4.2 Hardware and Strict Wall-Clock Execution

All experiments were conducted on reference hardware comprising an AMD/Intel x86_64 host with an NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM, driver version 580.173.02) under Linux kernel 7.1. Time limits were enforced strictly using monotonic CPU/GPU timers (`time.perf_counter()`). Simulated or synthetic throughput models were strictly prohibited. The budgets actually executed were 0.5 s and 1.5 s per trial in the budget matrix (P154) and 0.4 s and 0.8 s per trial in the paired baseline campaign (P155); the 10 s–1 h budgets of the original preregistration were not executed and are not claimed. Peak VRAM allocation stayed below 3 MB in every recorded run, against a 6 GB budget.

Retraction notice (D040, P161). Table 2 no longer shows the pre-retraction P158 verdicts: it records the **audited** post-retraction values. The original overall **SUPPORTED** verdict of P158

Table 2: Audited empirical verification status across all five falsifiable criteria of Hypothesis H1 (`docs/HYPOTHESIS.md`). Every metric is measured across independent seeds on reference hardware (NVIDIA RTX 4060 Laptop GPU). The P158 SUPPORTED verdict was retracted in P161 (D040); the verdicts below are the post-retraction audited values, consolidated statistically in P169.

Criterion	Scope	Observed Value	Preregistered Threshold	Verdict
C1: Rediscovery	Primary Level-A targets	80.0%	$\geq$ 80.0% in 4/5 seeds	**SUPPORTED**
C2: Speed & Cascade	Full cascade vs. naive scoring	15.51×	$\geq$ 10.0× speedup	**SUPPORTED**
	Early termination	3.00×	$\geq$ 2.0× speedup	**SUPPORTED**
C3: Evolution Value	Paired Wilcoxon signed-rank	$W = 55.0$, $p_{\text{Holm}} = 2.49 \times 10^{-3}$	$p_{\text{Holm}} \leq 0.01$	**SUPPORTED**
C4: Generalization	Extrapolation split (X_{extrap}), EvoByte	80.0%	$\geq$ 70.0% accuracy	**SUPPORTED**
	Asymptotic explosion	0.0%	0.0% explosion rate	**SUPPORTED**
	Average model length	5.2 opcodes	$\leq$ 25.0 opcodes	**SUPPORTED**
C5: Honest Baselines	Classical-SR success rate	**100%** vs. 80% (EvoByte)	EvoByte $\geq$ honest baselines	**NOT SUPPORTED**
	Classical-SR hidden error	3.913×10^{-13} vs. 0.9787	EvoByte $\leq$ honest baselines	**NOT SUPPORTED**
	PySR adapter	0/20 runs (not executed)	measured comparison	**NOT RUN**
Overall Hypothesis Verdict (audited):				**NOT SUPPORTED**

Table 3: Strictly paired empirical comparison of EvoByte vs. Random Search under identical wall-clock timeouts and hardware seeds (20 paired trials).

Search Method	Mean MSE	95% Bootstrap CI	Wilcoxon W	Raw p-value	Holm-Adjusted p
EvoByte (Ours)	**0.9787**	[0.1724, 1.9573]	**55.0**	**2.49**$\times 10^{-3}$	**2.49**$\times 10^{-3}$ (≤ 0.01)
Random Search	36.4172	[19.2030, 53.9985]			

is retracted because the honest-baseline criterion C5 is not met (the classical symbolic-regression baseline dominates EvoByte and the PySR adapter was never executed). The authoritative verdict of this paper is **NOT SUPPORTED**. See §5.1.
The consolidated P169 statistical verdict is reported in Section 7.

5 Empirical Results & Statistical Verdict

5.1 Hypothesis H1 Status

Table 2 lists the preregistered thresholds next to the P158 verdicts, and Table 4 lists the paired baseline measurements that invalidate them. The measured facts are: EvoByte rediscovers the Level-A targets in 80% of trials (C1); the cascade and early termination reach 15.51× and 3.00× (C2); evolution beats random search at $p_{\text{Holm}} = 2.49 \times 10^{-3}$ (C3); EvoByte reaches 80% extrapolation success with 0% explosion and 5.2 opcodes on average (C4); and the classical symbolic-regression baseline reaches 100% success with lower hidden error and higher throughput than EvoByte, while the PySR adapter was never executed (C5). Consequently, H1's honest-baseline criterion is **not supported** and the overall verdict of this paper is **NOT SUPPORTED**.

5.2 Statistical Superiority of Evolutionary Search (C3)

To test whether evolutionary selection provides genuine optimization value over stochastic exploration under fixed wall-clock budgets, we conducted paired trials between EvoByte and Random Search across identical target functions, timeouts, and random seeds (20 paired trials).

As reported in Table 3, EvoByte achieves a mean MSE of **0.9787** (95% empirical bootstrap confidence interval: $[0.1724, 1.9573]$), compared to **36.4172** ($[19.2030, 53.9985]$) for Random Search. A non-parametric Wilcoxon signed-rank test yields test statistic $W = 55.0$ with a raw p-value of 2.4884×10^{-3}. Applying Bonferroni–Holm step-down adjustment for Family-Wise Error Rate control maintains:

$$p_{\text{Holm}} = 2.4884 \times 10^{-3} \leq 0.01 \tag{6}$$

Table 4: Measured paired baseline campaign (P155). 20 runs per method on two Level-A targets, five seeds, budgets 0.4 s and 0.8 s. The PySR adapter was not executed.

Method	Completed	Success	Mean hidden MSE	Median hidden MSE	Mean CVPS
Random	20/20	50%	36.4172	22.2844	4,205.0
Classic-GP	20/20	50%	11.0541	8.6860	44,561.9
Classical-SR	20/20	**100%**	3.913×10^{-13}	3.913×10^{-13}	26,116.5
PySR-Adapter	**0/20**	not run	not run	not run	not run
EvoByte (Ours)	20/20	80%	0.9787	3.197×10^{-14}	4,799.5

This establishes with statistical significance ($p < 0.01$) that evolutionary selection in compact bytecode provides substantial performance advantages over random exploration under identical wall-clock constraints.

5.3 Paired Baselines (C5)

Table 4 reports the paired baseline campaign: five methods, two Level-A targets, five seeds and two wall-clock budgets (0.4 s and 0.8 s), 20 runs per method, all on the same RTX 4060 Laptop GPU. The PySR adapter could not be executed because its dependency (Julia/PySR) is not installed in the measured environment; it is reported as *not run* and never counted as a win or as a zero-error result.

Two measurement caveats apply to Table 4 and are part of the reported result. First, the *Classical-SR* baseline is a least-squares polynomial fit over degrees 1–5; the two Level-A targets are themselves polynomials, so this baseline class covers the target class exactly and its 100% success rate reflects target-class coverage rather than a stronger search procedure. Second, the CVPS column is not computed on a common basis: EvoByte counts verified bytecode candidates, whereas the *Classical-SR* row divides five polynomial fits by the fit time alone. The comparison of throughputs is therefore indicative, while the comparison of success rate and hidden error is direct. On the direct comparison, the honest baseline is at least as good as EvoByte, and criterion C5 is not met.

5.4 Generalization and Anti-Bloat Properties (C4)

Traditional tree-based GP frequently minimizes training error by memorizing training points, resulting in massive polynomials that diverge wildly outside the training domain. To assess generalization, models discovered by EvoByte were evaluated on:

1. **Hidden In-Domain Split (X_{hidden}):** Randomly withheld points within the training support $[x_{\min}, x_{\max}]$.
2. **Extrapolation Split (X_{extrap}):** Out-of-domain points evaluated over $[x_{\max}, x_{\max} + \Delta]$.

EvoByte models achieved an 80.0% extrapolation rediscovery rate with a **0.0% asymptotic explosion rate**. The mean program length was **5.2 opcodes**, demonstrating that fixed-width bytecode acts as a natural structural regularizer that eliminates tree bloat by design. On the same hidden and extrapolation splits the *Classical-SR* baseline reached a 100% success rate, and the *Classic-GP* baseline 50%; the generalization numbers of EvoByte are therefore not a distinguishing advantage over the honest baselines.

5.5 Pareto Frontier Analysis (C5)

Table 5 details the empirical non-dominated Pareto frontier mapping bytecode complexity against out-of-sample generalization error (Hidden MSE). The frontier does *not* show EvoByte dominating the classical baseline: EvoByte's most compact point (3 opcodes, hidden MSE

Table 5: Empirical non-dominated Pareto frontier mapping program complexity (byte/opcode count) against out-of-sample generalization error (Hidden MSE).

Pareto Point	Discovered By	Benchmark Target	Opcode Count	Hidden MSE
P_1	Random Search	$y = x + 1$	2	0.000×10^{0}
P_2	Random Search	$y = x^2 + 3x + 7$	2	4.900×10^{1}
P_3	EvoByte	$y = x^2 + 3x + 7$	3	1.773×10^{-12}
P_4	Classical-SR	$y = x^2 + 3x + 7$	6	7.796×10^{-13}
P_5	EvoByte	$y = x^2 + 3x + 7$	8	6.395×10^{-14}

1.773×10^{-12}) is smaller than the *Classical-SR* point (6 opcodes, 7.796×10^{-13}), which attains the lower hidden error, and a *Random* point with 2 opcodes attains zero hidden error on $y = x+1$. EvoByte occupies part of the frontier, but the honest baselines occupy it as well.

6 Architectural Ablation Studies & Negative Results

To understand the exact contribution of each subsystem, we evaluated nine isolated variants (Table 1).

6.1 Verification and Acceleration Components

The cascade architecture demonstrated measurable efficiencies: the complete cascade against naive full-dataset scoring provides a 15.51× speedup, early termination a 3.00× speedup, and GPU instead of CPU execution of the same bytecode a 2.60× speedup (Table 1, P157 battery, single target $y = x^2 + 3x + 7$).

6.2 Negative Result: Neural Micro-Generator Rejection (ADR-0010)

Phase P12 investigated augmenting evolutionary search with a lightweight neural proposal network trained to predict promising bytecode tokens. Empirical measurement revealed that forward inference through the neural network introduced substantial overhead, reducing overall candidate evaluation throughput from $> 2,000$ CVPS to ≈ 760 CVPS (a 0.38× slowdown). Because the quality improvement per proposal failed to compensate for the 2.6× reduction in evaluated candidates per wall-clock second, the neural generator was formally rejected and dropped from the architecture under Architectural Decision Record ADR-0010. We report this negative result to emphasize that in ultra-fast verification regimes, raw throughput and simple stochastic mutation frequently outperform neural guidance.

7 Validated Discovery Program (P161–P170)

After the P160 manuscript, every published claim was mechanically re-audited and the research program continued under preregistration. The audited record is:

- **Claim retraction (P161, D040).** A mechanical integrality audit of the 58 generated Schinzel identities re-evaluated each identity exactly in $\mathbb{Q}[k]$: only 9 produce positive integers for all k; 49 admit only rational solutions (in $\mathbb{Q}$ every fraction decomposes as a unit fraction, so those 49 are no evidence). The `novel-extension` classification was retracted.

- **Lean statement fidelity (P162).** A conservative classifier audited all 4888 generated Lean theorems: 51 are provably vacuous constants, 2012 are numeric literals, and 2825

Table 6: Preregistered record-search campaign (P166): 120 trials (3 targets × 2 sizes × 2 algorithms × 10 seeds, 10 s each), certified by the independent P163 verifiers. `Tie` = published bound, `beat` = bound +1.

Target	Size	Trials	Certified	Best violations	Verdict
`vanderwaerden_w3_3`	tie (26)	20	10	0	**DISCOVERY_TIED**
`vanderwaerden_w3_3`	beat (27)	20	0	1	
`vanderwaerden_w2_6`	tie (1131)	20	0	2,303	**NULL_RESULT**
`vanderwaerden_w2_6`	beat (1132)	20	0	2,285	
`schur_s5`	tie (160)	20	0	4	**NULL_RESULT**
`schur_s5`	beat (161)	20	0	7	

need human review (a lower bound on vacuity, never a certificate of meaning). A human review registry gates any future claim of a result.

- **Record-target catalogue (P163).** Four frozen extremal targets with published bounds and independent polynomial-time certificate verifiers in pure Python and NumPy.
- **Honest combinatorial baselines (P164).** Fully implemented Random Search, Simulated Annealing and Tabu Search under paired wall-clock budgets (0.5 s, two seeds): best 6, 1 and 1 monochromatic violations on [1..27]; no certified solution.
- **Blind rediscovery (P165).** Tabu Search re-found the published $W(3,3) = 27$ optimum (valid 3-coloring of [1..26]) in 4/5 seeds within 10 s (mean 6.26 s); Simulated Annealing found none. Independent exhaustive search confirmed $W(3,3) = 27$ and $W(2,4) = 35$, and reproduction of random runs confirmed eligibility.
- **Preregistered record campaign (P166).** Table 6 reports the frozen-manifest campaign: 120 trials, 12,016,204 real evaluations, 1,134.1 s of machine time. One target tied the published record (10 certified colorings of [1..26]); two ended as **null results** (`NULL_RESULT`). No baseline beat a published bound; the best near-miss on [1..27] had exactly 1 violation.
- **External validation (P167).** A standalone verifier using only the Python standard library certifies exported certificates in under 1 s; the sample certificate verifies end-to-end (SHA-256 `b77e1f59...`).
- **Faithful Lean certificates (P168).** Two kernel-verified theorems (¬ monochromatic 3-AP on [1..8], ¬ Schur violation on [1..4]), compiled by `lake build` with zero `sorry`, passing the P162 linter with 0 `vacuous-constant`, and recorded with explicit human review.
- **Consolidated statistical verdict (P169).** Six paired method comparisons (Simulated Annealing vs. Tabu Search, matched by target, size and seed) are all non-significant after Holm FWER control at $\alpha = 0.01$; the pooled $N = 60$ comparison favors Simulated Annealing ($W = 285.0$, $p_{\text{Holm}} = 0.00108$). Mathematical record discovery is **NULL_RESULT** and bytecode evolutionary advantage is **NOT_SUPPORTED** (the campaign has no EvoByte arm by design).

8 Reproducibility & Open Science

To guarantee independent reproducibility, EvoByte provides:

1. **Hermetic Containerization:** A Dockerfile based on `python:3.11-slim` that installs all runtime dependencies and executes the verification suite under `--network none`, ensuring zero network leaks or unpinned dependencies.

2. **Open Research Data Bundle:** A complete open-access research archive formatted for Zenodo and OSF deposit (`scripts/export_zenodo_bundle.py`). Exported from the audited tree (544 files, 5.46 MB), the package contains the manuscript and PDF, raw hardware execution logs, JSON telemetry, formal Lean 4 certificates, the P161–P170 evidence artifacts, the project source code and the standalone external verifier, validated with 100% SHA-256 cryptographic checksums via `manifest-sha256.json`; the top-level `final-manifest-sha256.json` records the bundle, tarball and per-artifact hashes. The file count follows the exported tree and is regenerated by the export script.

9 Related Work

Tree-Based Genetic Programming: Pioneered by Koza [5] and Poli et al. [9], tree GP remains widely used. Systems such as gplearn and Operon provide optimized CPU implementations but remain constrained by variable tree bloat.

GPU Acceleration in Evolutionary Computation: Early works by Langdon and Banzhaf [6] demonstrated SIMD interpreters on GPUs. EvoByte extends linear GP principles [7] by integrating multi-tier deterministic verification cascades directly in VRAM.

Physics-Inspired and Modern Symbolic Regression: AI Feynman [11] combines dimensional analysis with neural networks. PySR [2] utilizes multi-population evolutionary algorithms in Julia. EvoByte complements these approaches with a bytecode engine that verified between 500 and 25,017 candidates per run under the sub-second budgets measured here; a direct comparison against PySR was not performed because the dependency was unavailable.

10 Conclusion

We have presented **EvoByte**, an open-source, massively parallel symbolic discovery architecture based on fixed 64-byte linear bytecode and hierarchical verification cascades. Through multi-seed wall-clock experiments we measured that evolutionary search over compact bytecode with fast verification outperforms random search per unit of compute time ($W = 55.0$, $p_{\text{Holm}} = 2.49 \times 10^{-3}$), and that the verification cascade and early termination raise measured throughput by 15.51× and 3.00×. We also measured that a classical symbolic-regression baseline reaches a higher success rate (100% versus 80%) with lower hidden error than EvoByte, and that the PySR adapter could not be executed in our environment. The honest-baseline criterion of Hypothesis H1 is therefore **not supported**.

The P161–P170 validated-discovery program re-audited every claim of this paper: a mechanical integrality guard retracted 49 Schinzel novelty claims, a Lean fidelity gate bounded the provably vacuous theorems (2063 of 4888), a preregistered 120-trial combinatorial campaign (12,016,204 evaluations) found no new mathematical record (one tied published record, two null results), and the consolidated statistical verdict confirms H1 **not supported** with **null result** for mathematical record discovery. Two faithful Lean 4 certificates are recorded with explicit human review, and a standard-library-only external verifier reproduces the sample certificate end to end. We report these negative results together with the complete open research data (SHA-256 chain of custody), the hermetic container recipe, and the retraction notices that accompany the previously published verdicts.

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